Genealogy of the Programme

A Lakatosian rational reconstruction

Patrick S. Mahon

Written 2026-06-06

Abstract

A Lakatosian rational reconstruction — the actual sequence of conjectures, refutations, and problem-shifts across the full 631-commit history (2024-09 → 2026-06), recovered so the shape of the whole is visible from outside the turn-to-turn. This document exists because the programme spent two months contracting (three papers + four active leads in early May → one open question now) and the largest-scale shape was no longer visible from inside it. It is a map, not a verdict. It deliberately leaves one question open at the end — the one that decides what the next year is for. See `program_overview.md` for the programme's own statement of its central question, and `knowledge_map/` for current lead state.

Contents

1 The one-sentence shape

It began as applied operator theory (compute the resolvent of a nonlinear semigroup from data), drilled downward through three layers of foundation until it hit the origin of the σ-algebra itself, discovered the bedrock result there (CE), watched that result get reabsorbed into something classical, and has been pruning back ever since toward the one genuinely novel survivor (point-free σ-additive probability on non-distributive lattices). The repository name, "Resolvent_Framework," is a fossil of a question the programme no longer asks.

2 The four eras

2.1 Era 1 (2024-09 → 2025-10) — Operators from data

The founding question: given a trajectory ensemble $\{X_t\}$, build a nonlinear semigroup, recover its generator, characterize its resolvent. Real early result: the resolvent set = complement of the union of finite-difference spectra. Then a reframing pivot (not a wall) — the **Rose operator**, a two-dial homotopy between Koopman and Perron–Frobenius, plus information geometry (Fisher / Onsager–Machlup). The constant was always *operators sitting between determinism and noise, fit from data.* The literal resolvent computation quietly receded. The repository name fossilizes here.

2.2 Era 2 (2026-03) — The descent into foundations

After a 5-month incubation on a side branch, the **query system** abstraction lands fully-formed on 03-18: a dynamical system is a directed family of finite observations. Once observation is primitive, the question stops being "what is the operator?" and becomes "when does a compatible family of finite observational laws assemble into a single σ-additive probability?" The programme drills down: Paper 4 (delay/dynamics, Takens as corollary) → Paper 0 (Prokhorov: σ-additivity derived) → Paper -1 (discriminability: where does the σ-algebra come from?). The famous line: *"we thought we were climbing a mountain; we had reached the valley floor."* The join at the bottom is **CE (collective exhaustion)** — characterizing exactly when finite additivity extends σ-additively — reframed via Stone duality.

2.3 Era 3 (2026-04, the 310-commit peak) — Build-out and the first kills

Everything consolidates into three papers (Probability from Observation / Dynamics from Probability / Reconstruction from Observation), spawns Paper IV (finite-sample reconstruction, the $\delta(L)$ separation defect), spawns the bridge note, spawns a standalone logic paper (*Countable Additivity is Not First-Order Axiomatizable*) which spawns its own ZFC-independence ladder. Then the turn to self-criticism: the honest "mathematics-complete $\neq$ submission-ready" correction, four-papers-collapse-to-three, the first mis-citation kills, the fibre-mixing irreducibility thread downgraded when its analogy to CE broke.

2.4 Era 4 (2026-05 → 06) — The pruning

The institutional event: on **2026-05-11, CE collapses** — demoted from a novel condition to "notation for σ-additivity." With CE gone as load-bearing, the dependent papers had nothing under them, and Papers II and III were withdrawn in 48 hours ("classical ergodic theory repackaged"). The audit pipeline (`active_leads` / `covered_leads` / `knowledge_map`, "audit before drafting") is scar tissue from that loss. Then systematic pruning: fibre mixing (killed — bridge theorem false), coherence/CE-completion (parked — Howson 2008 got there first), tail-decay (parked — SSK independence), κ_Q (parked), epistemic-boundary unification (parked). The OML thread, seeded from Paper I's closing move, becomes the sole survivor — splits into **extension axis** (now closed on $L(H)$) and **descent axis** (the one open question), reframed as point-free σ-additive probability.[1]

[1] Descent-axis status *since* closed/parked (2026-06-10, `/audit full` on concreteness): $\mathcal{L}_{\mathrm{MO}_2}$ is concrete (conjecture false as worded) but trivial and already characterized by Pták–Pulmannová 1994; no contribution. This document narrates history, not live state — see `program_overview.md` (item 4, authoritative) and the parked map `notes/covered_leads/descent_axis_residue_post_kill.md`.

3 The thread map (splits, joins, deaths)

```
RESOLVENT of nonlinear semigroup (2024-09) -- the founding question
   |  Hille operator, Cn-semigroups, finite-difference spectra
   v
ROSE OPERATOR  (2025-06, reframe) -- Koopman <=> Perron-Frobenius homotopy
   |  + information geometry (Fisher / Onsager-Machlup)
   v
QUERY SYSTEM  (2026-03-18, the unlock) -- dynamics = directed observations
   |
   |--> delay/Takens ---------------> [Paper III reconstruction] --+
   |--> Prokhorov (s-add. derived) -> folded into Paper I          |
   +--> discriminability -----------> CE THEOREM <-- the valley floor
                                        |  (Stone duality reframe)
                                        |
            +--------------------------+--------------------------+
            v                          v                            v
      [Paper I: CE/Stone]       [logic note: CE not          [Papers II/III:
            |                    first-order] --> ZFC           dynamics + recon]
            |                    ladder (parked,                    |
            |                    Strategy D independent)            |
            |                                                       |
   2026-05-11: CE COLLAPSES --> "notation for s-additivity"         |
            |                                                       v
            |                                          II & III WITHDRAWN
            |                                          ("classical, repackaged")
            v                                                       |
   Paper I survives                           +--- delay/recon reopened as
   (ultrafilter geometrization)              |     Phase-1 seed -> PARKED
            |                                |     rival-realiz -> PARKED
            v                                +---- distributed-sensor -> PARKED
   OML THREAD (seeded from Paper I's close)
            |
            |- EXTENSION axis --> CLOSED on L(H) (no state extends, n=4) [Type 5]
            +- DESCENT axis ----> THE ONE OPEN QUESTION (as of writing)
                                   (Navara-over-MO2 s-orthocompleteness check)
                                   = point-free s-additive probability / II PR
```

(For the current descent-axis status — now CLOSED/PARKED after the `/audit full` *on concreteness (2026-06-10):* $\mathcal{L}_{\mathrm{MO}_2}$ *concrete but trivial + already characterized (Pták–Pulmannová 1994), no contribution — see* `program_overview.md` *(item 4, authoritative) and the parked single-source map* `notes/covered_leads/descent_axis_residue_post_kill.md`*. This document narrates history, not live state.)*

4 What this says about the "loss at the largest scale"

Three observations, named rather than soothed:

1. **The programme has been getting smaller for two months, and that is the system working, not failing.** Every contraction was an honest kill — a primary source read, a counterexample built, an audit run. The loss is real subtraction, but the thing doing the subtracting was the user's own tooling discipline (verify-by-building, audit-before-drafting). A programme that prunes itself this cleanly is rare.
2. **The central trauma already happened — and was survived.** The CE collapse on 05-11 is the real loss event; CE was the keystone the whole "valley floor" descent was for.

Everything since is aftershock — the programme confirming the blast radius. The recent "ooof" about probabilistic realism is delayed grief for news already processed locally, commit by commit, without ever stepping back to feel the whole.

3. **There is a genuine through-line, and it survived.** Strip every name and the constant from 2024 to now is: *structure that observation alone forces, without assuming the space you're reconstructing into.* But see the next section — this line has a crack in it, and the crack is the decision.

5 The open question: which curiosity is driving (the (a)/(b) fork)

This is the part the genealogy cannot settle, and the part that decides whether the descent check is the right next year. It is left open deliberately.

5.1 The crack in "stripped to its bones"

The through-line had four instances. **Three are Boolean. One isn't.** Resolvent-from-data, σ-additivity-derived, reconstruction-without-manifold — all classical, all distributive. The descent question is the only non-distributive one. The "against assuming the target space" framing groups all four together — but the survey itself (descent survey, lines 544–550) says the descent question's novelty is *point-free* $\times$ *σ-additive* $\times$ *natively non-distributive* $\times$ *directed-built*, and **the point-free / no-assumed-space part is not the novel part.** Localic measure theory already does point-free σ-additive probability on distributive frames (Vickers, Simpson, Coquand–Spitters — survey lines 535–538). So:

> **The entire novel content of the descent question is non-distributivity — handling mutually incompatible observations. So the descent question serves curiosity (b), not (a). If the driving curiosity is (a) — "don't assume the space, derive it from coherence" — then CE / Paper I *already delivered it* (Boolean case), and the descent question is a quantum-foundations question that grew in late and got mistaken for the origin because it's what survived the pruning.**

This is the load-bearing sentence. The doc's job is to record it so it can't be un-seen, then leave the choice open.

5.2 The two curiosities, made vivid

Researcher (a) — the anti-smuggler. What bothers them is positing the answer to reason backward. Takens assumes the manifold; Kolmogorov assumes the sample space. The 2024–2026 descent was them saying "no — derive what observation alone forces." Satisfaction = watching structure emerge from coherence without being snuck in. For (a) the deep result is **CE / Paper I** (coherence forces a unique σ-additive measure, no Ω). *It's delivered.*

Researcher (b) — the incompatibility theorist. What grips them is that some observations can't be made together, and classical probability has no room for it. Satisfaction = understanding the structure of incompatibility itself: when it forces you out of the Boolean world, what survives, what "probability" means on a non-distributive logic. For (b) the deep result is **the descent question** (can probability live on the entailment relation of incompatible propositions at all?). *Nothing here is done.*

The discriminating test — which sentence makes you lean in:

- (a) "You don't need to assume a state space — observation forces the probability."

- (b) "When observations can't be jointly performed, the logic of events stops being Boolean — and here's what happens to probability."

5.3 What the fork changes

	If you're (a)	**If you're (b)**
Is the descent question right?	**No** — harder variant whose novel part (non-distrib.) is incidental	**Yes** — non-distributivity *is* the whole novel content
What's already done?	Paper I delivered it (Boolean). You'd polish a closed thing.	Nothing — point-free σ-add. on non-distrib. lattices is vacant
Still-open relative	Strategy D / ZFC-independence (more continuous with 2024 origin)	The descent check itself — one derivation from go/park

5.4 How to feel which one you are (sit with these; don't answer fast)

1. **When did you feel most alive?** The moment CE clicked (coherence forces the measure, no Ω) $\rightarrow$ (a). Or when the observation algebra broke Boolean and you reached for OMLs $\rightarrow$ (b)?
2. **The skew-product.** Filed this session as a limitation (invisible fibre, space unrecoverable). To (a) it's a melancholy fact — the space you wanted is unrecoverable. To (b) it's not even interesting — still Boolean, still co-realizable. Which was your gut?
3. **The quantum connection.** Paper II is quantum foundations (Kochen–Specker, Gleason, realism). Does it thrill you (arrived somewhere deep) $\rightarrow$ (b)? Or feel like a detour from a question never about QM $\rightarrow$ (a)?

5.5 The honest finding

The genealogy is consistent with **both** readings. The origin was (a)-flavored; the survivor serves (b). Curiosities are allowed to evolve from one to the other, and (b) is a perfectly good thing to have grown into. But the choice should be made knowing it *is* a choice — not let the momentum, the sunk cost, or the fact that the descent question is "the last thread standing" decide it.

Status: open. No rush. This is genuinely yours to settle.

Update 2026-06-12 — evidence, not a verdict. Working the descent problem, the user chose **Reading 1** for its success criterion (relational = no hidden realisation; the (a)-flavored answer *for that problem*). Two consequences bear on the deeper fork without closing it: (i) the descent prize, taken (a)-wards, *reduced to a contextuality / compactness-failure question that reconnects to CE / Paper I* — the (b)-looking survivor rejoined the (a)-origin; (ii) Strategy D (purely (a)) advanced the same session. Both faces now point back toward (a). Suggestive that the driving curiosity was (a) all along — but choosing Reading 1 for a problem is not self-diagnosing the curiosity, so the fork stays **open** by this document's own standard.

Update 2026-06-18 — Strategy D KILLED on prior-art. The worked math — $\mathrm{Clop}(Y, \mathfrak{T})$ (Argyros pre-Gleason) is **not** σ-complete (gap family $A_k = [0^{k-1}11]$ at $\bar{0}$, no least clopen upper bound) — is true but clears no contribution bar (`/audit full`): **Gaifman 1964** already inhabits the ZFC cell, *stronger* (atomless BA, no strictly-positive *finitely*-additive measure);

non-σ-complete + atomless are trivial, measure-freeness is Argyros's published theorem; "Strategy D" was a private name, never field-open. So the (a)/anti-smuggler face has **one** delivered result (CE/Paper I); the (b)/incompatibility face (OML σ-essential descent) is the lone open thread. The "Still-open relative = Strategy D" row above is superseded (the relative is *retired*, not solved). This removes the (a)-side's last open target — continuing OML descent is now an explicitly (b)-committed choice — but does not settle the fork.